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Fig. 3: Author's prototype assembled on general-purpose PCB

AC mains supply is available. When RL1 is energised, its common point (pole) is connected to NO point.

When AC mains supply stops, the relay's common point (pole) gets connected to NC point. This completes the connection of the load (bulb) via transformer X2, which is driven by IRF250 power MOSFET and 2N2222 transistors based circuitry to generate AC from transformer X2.

So, working of the circuit is simple. Transformer X1 is used to drive the relay when 230V AC mains

supply is available to power the load (a bulb in this case). And when this supply goes off, transformer X2 is used to power the load via its primary—while its secondary is connected to the battery and the oscillator circuit. Thus, the bulb always remains on.

Construction and testing

The circuit can be assembled on a general-purpose PCB and enclosed in a suitable box. Connect 230V AC mains supply to primary terminals of transformer X1 across CON1. Connect the load (bulb) across CON2. The circuit is now ready to use. The author's prototype assembled on a general-purpose PCB is shown in Fig. 3.

While assembling the circuit, use heavy-duty wires between the battery and transformer X2, and between the load and MOSFET, as they will be carrying higher current. Also, use heatsinks for the two MOSFETs.

EFY note. The design does not have a charger circuit, so ensure that the battery is fully charged before it is used. **EFY**

PARTS LIST

Semiconductors:

IC1	- LM7805, 5V voltage regulator
D1	- 1N4007 rectifier diode
BR1	- 1A, bridge rectifier
IRF1, IRF2	- IRF250 MOSFET
T1, T2	- 2N2222 NPN transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R4	- 650-ohm
R2, R3	- 12-kilo-ohm

Capacitors:

C1	- 1000 μ F, 50V electrolytic
C2, C3	- 100nF ceramic disc
C4, C5	- 2.2 μ F, 50V electrolytic

Miscellaneous:

CON1-CON2	- 2-pin connector
RL1 relay	- 5V, single changeover relay
BATT.1	- 12V, 7Ah battery
X1	- 230V AC primary to 9V, 500mA secondary transformer
X2	- 230V AC primary to 12-0-12V, 5A secondary transformer
	- General-purpose PCB/Breadboard
	- 230V, 25W bulb (load)
	- Heatsinks for IRF250

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